

$$\sum_{n=1}^{\infty} \frac{(3-n)(n^2-16)^2}{(n+1)(5n^2-6)^3}$$

What conclusion can be reached by using the n^{th} term test on the series?

Choose 1 answer:



INCORRECT

The series diverges.



INCORRECT

The series converges.



CORRECT (SELECTED)

The test is inconclusive.

1 / 2

The n th term test says the following:

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=0}^{\infty} a_n$ diverges.

If $\lim_{n \rightarrow \infty} a_n = 0$, then $\sum_{n=0}^{\infty} a_n$ might converge or might diverge; i.e., the test is inconclusive.

2 / 2

We see that $\lim_{n \rightarrow \infty} \frac{(3-n)(n^2-16)^2}{(n+1)(5n^2-6)^3} = 0$ because the degree of the denominator is greater than the degree of the numerator.

Therefore, the n th term test is inconclusive.

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n th term divergence test

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